Appendix 3 *for:* Beyond the average phenotype: individual variation alters mechanistic forecasts of climate responses

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Additional Results

1 Jensen bias for males

Figure S3.1 shows the spatial distribution of Jensen bias (J) between population-average and individual-level prediction surfaces for males under current European climate. While the population-average DEB parameterisation underestimated most performance metrics for females (**Fig. 2**), the pattern for males was largely opposite: the population-average parameterisation overestimated most metrics, including reproductive energy ($|J| = 2315$ J), basking time ($|J| = 286$ h) and foraging time ($|J| = 398$ h). In relative terms, male bias was largest for reproductive energy (normalised $|J| = 9.1\%$), R_0 (7.4%) and food intake (6.9%). Overestimation dominated for body size, reserve, food, reproductive energy and basking metrics ($J > 0$ in 57–96% of cells), whereas maximum and minimum body temperature showed predominantly negative bias ($J < 0$ in 84–93% of cells). The magnitude and direction of Jensen bias therefore depended strongly on both sex and ecological performance metric.

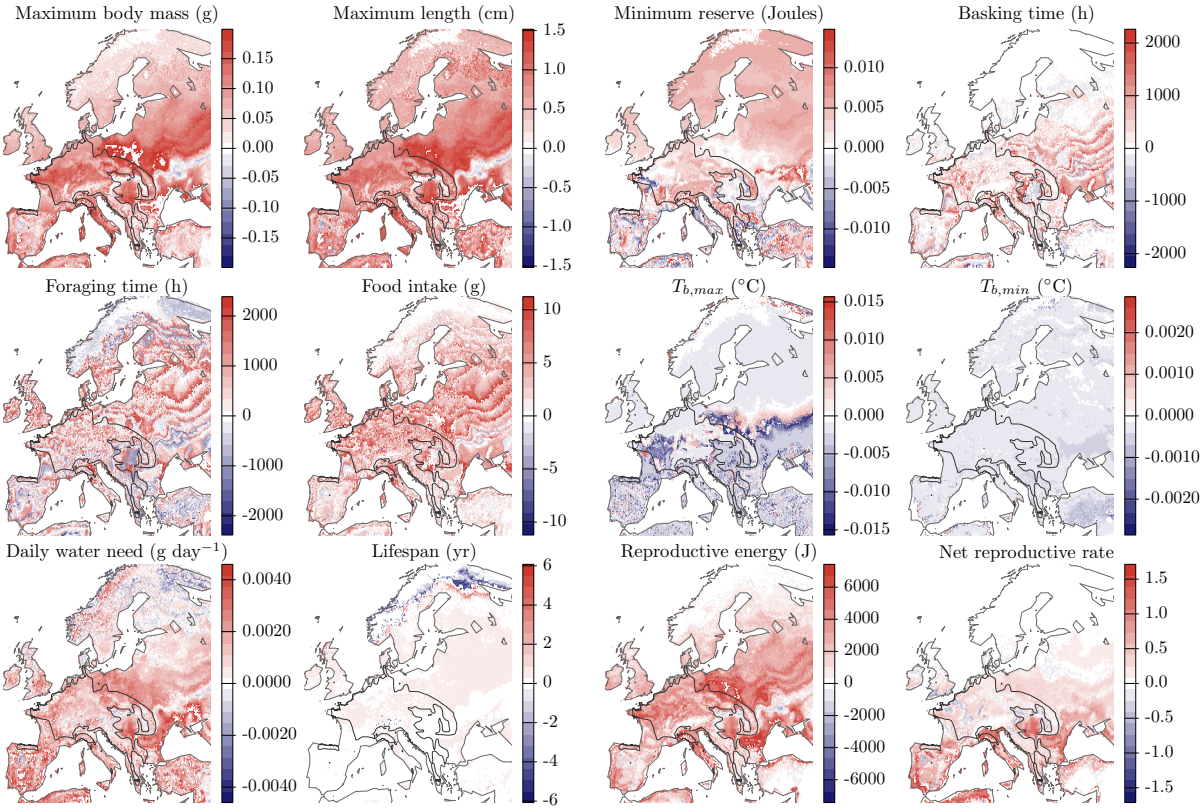


Figure S3.1: Jensen bias (J) between population-average and individual-level prediction surfaces for males. Values close to zero indicate close agreement; departures from zero indicate systematic differences in the spatial pattern. The colour scale is symmetric and centred at zero. The black outline delimits the current distribution of *I. alpestris* (IUCN 2021). T_b, body temperature.

2 Coefficient of variation maps

The coefficient of variation (CV) across the 37 individual-level predictions quantifies the degree of physiological heterogeneity independent of Jensen bias. High CV values identify regions and metrics where individual variation most strongly influences predicted outcomes.

2.1 Females

For females, CV was highest toward the range margins for most performance metrics (**Fig. S3.2**), indicating that the ecological consequences of individual physiological variation were spatially structured rather than uniform across the landscape.

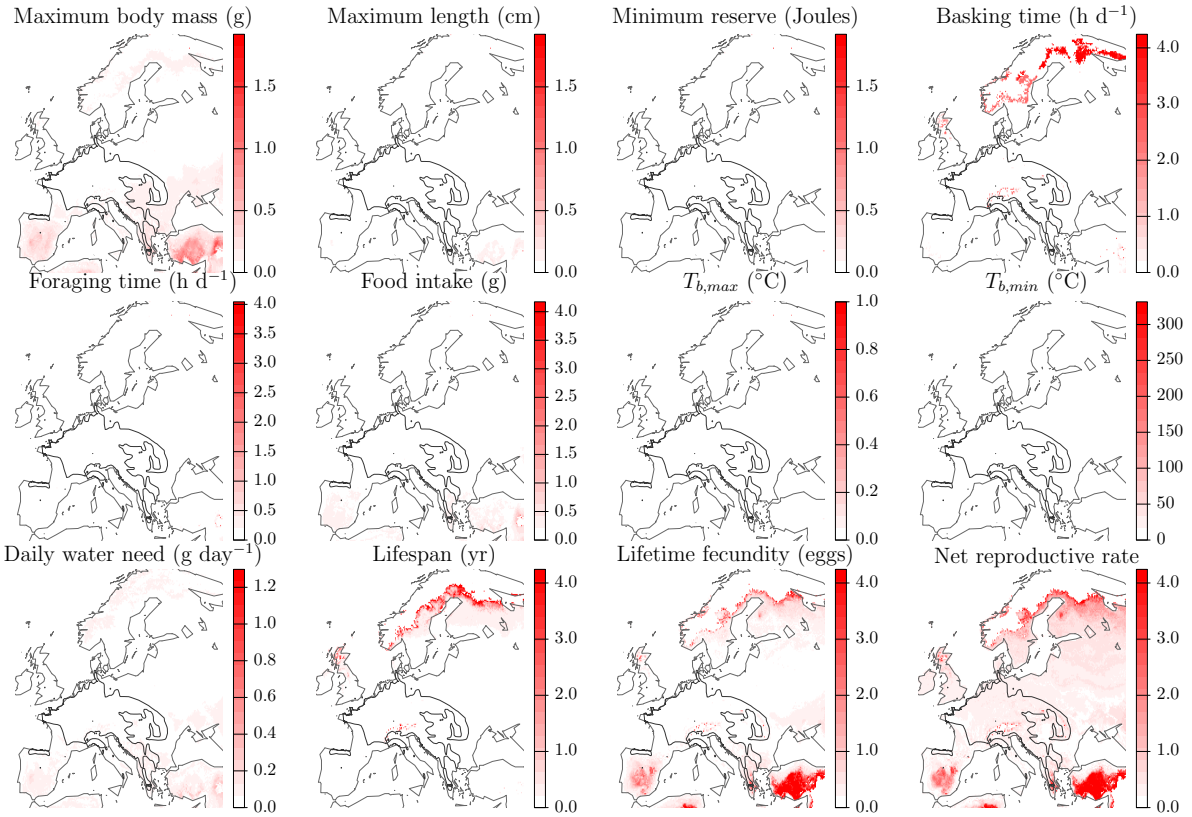


Figure S3.2: Coefficient of variation (CV) across individual-level predictions for females. Each panel shows the spatial distribution of CV for one performance metric. Higher values indicate greater individual-level variation in that metric across the geographic range. The black outline delimits the current distribution of *I. alpestris* (IUCN 2021). T_b, body temperature.

2.2 Males

A similar spatial pattern was observed for males (**Fig. S3.3**), with CV values peaking at range margins for most metrics, consistent with the pattern observed in females.

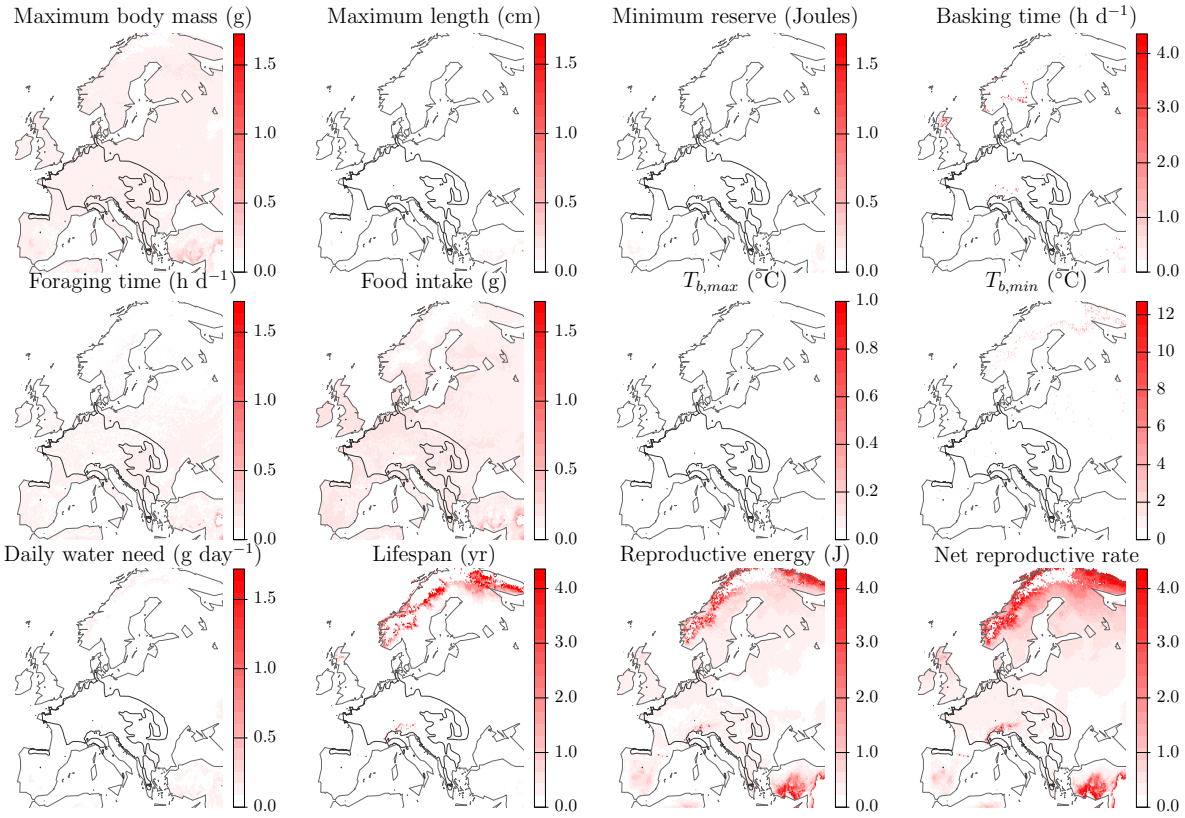


Figure S3.3: Coefficient of variation (CV) across individual-level predictions for males. Each panel shows the spatial distribution of CV for one performance metric. Higher values indicate greater individual-level variation in that metric across the geographic range. The black outline delimits the current distribution of *I. alpestris* (IUCN 2021). T_b, body temperature.

3 Climate warming reshapes Jensen bias (males)

Climate warming altered the magnitude of Jensen bias for all twelve male performance metrics (**Fig. S3.4**), mirroring the pattern observed for females (**Fig. 3**). However, as for females, warming did not produce a uniform response: the strongest relative amplification occurred for minimum body temperature (rel. $\Delta|J| = +510\%$; 60% of cells) and maximum body temperature (rel. $\Delta|J| = +215\%$; 62% of cells), whereas lifespan (-1%) was attenuated. The sign of J changed in 5–46% of cells under $+4\,^{\circ}\mathrm{C}$ warming, with highest rates for basking time (46%), foraging time (43%) and minimum reserve (31%). Even under $+2\,^{\circ}\mathrm{C}$ warming, the direction of change was already discernible for most metrics: 11 metrics showed increased $|J|$, with relative changes ranging from +10% (maximum body length) to +126% (minimum body temperature), while 1 metric was attenuated (minimum reserve (-1%)), and 1 metric (lifespan) switched from amplification at $+2\,^{\circ}\mathrm{C}$ to attenuation at $+4\,^{\circ}\mathrm{C}$, suggesting a nonlinear threshold in the averaging bias. For comparison, in females 1 metric was attenuated under $+2\,^{\circ}\mathrm{C}$ (minimum reserve (-5%)).

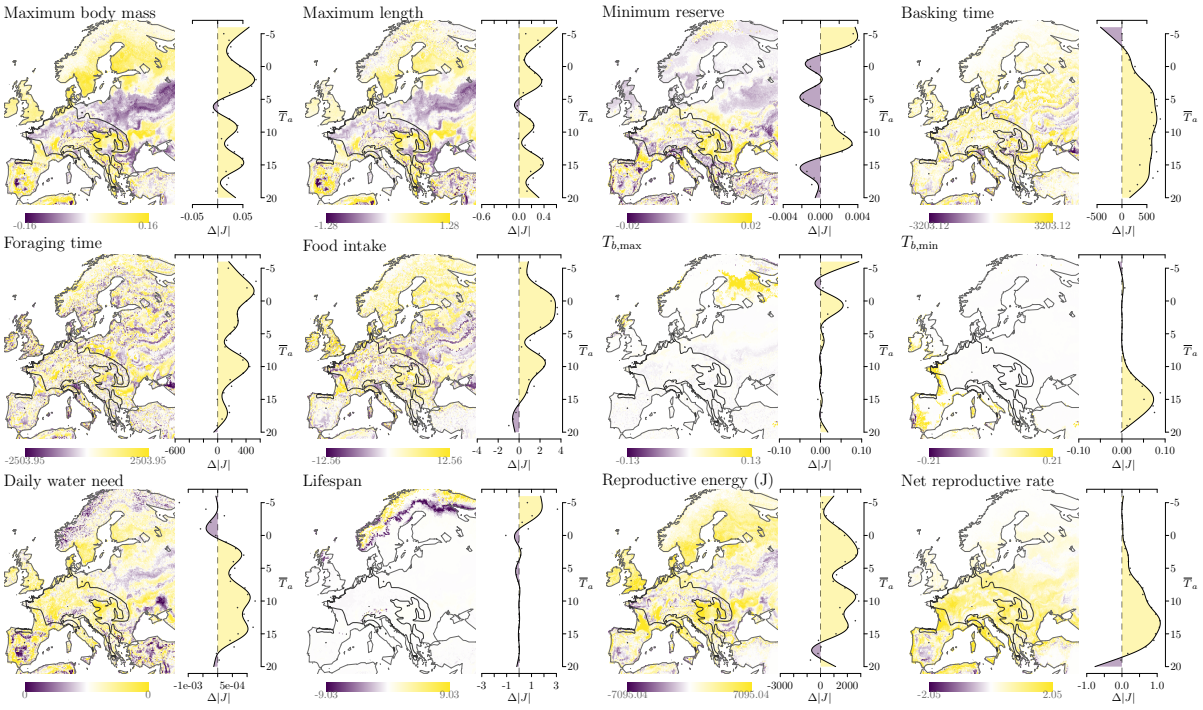


Figure S3.4: Jensen bias amplification under $+4\,^{\circ}\mathrm{C}$ warming for males. Each map displays the change in absolute Jensen bias ($\Delta|J|$, left, with colour bar) for each performance metric. The curve (descriptive GAM smoother) at the right of each map represents how $\Delta|J|$ changes along the mean annual temperature ($\overline{T}_a$) gradient. *Yellow* shading (positive $\Delta|J|$) indicates amplification of the averaging bias under warming, while *purple* shading (negative $\Delta|J|$) indicates attenuation. The black outline delimits the current distribution of *I. alpestris* (IUCN 2021). T_b, body temperature.

4 Consequences for ecological forecasts (males)

Jensen bias produced trait-specific differences between population-average and individual-level forecasts for males under both current and future climates (**Fig. S3.5**), mirroring the female pattern (**Fig. 4**). Reproductive energy exhibited the largest discrepancies, with the population-average model generally overpredicting relative to the individual-level mean under the current climate. The population-average model also overpredicted lifespan (84% of cells; median relative bias +2.1%) and maximum body mass (95% of cells; median +5.5%), and the median relative bias in reproductive energy under +4 °C was +10.9%. Lifespan showed a consistent tendency for the population-average model to predict higher values than the individual-level mean across climate scenarios, whereas maximum body mass showed comparatively close agreement, indicating relatively weak consequences of physiological averaging for this trait. The relative bias maps (**Figs. S3.6, S3.7**) express the same comparison as a percentage of the individual-level mean, making the spatial structure of the disagreement visible.

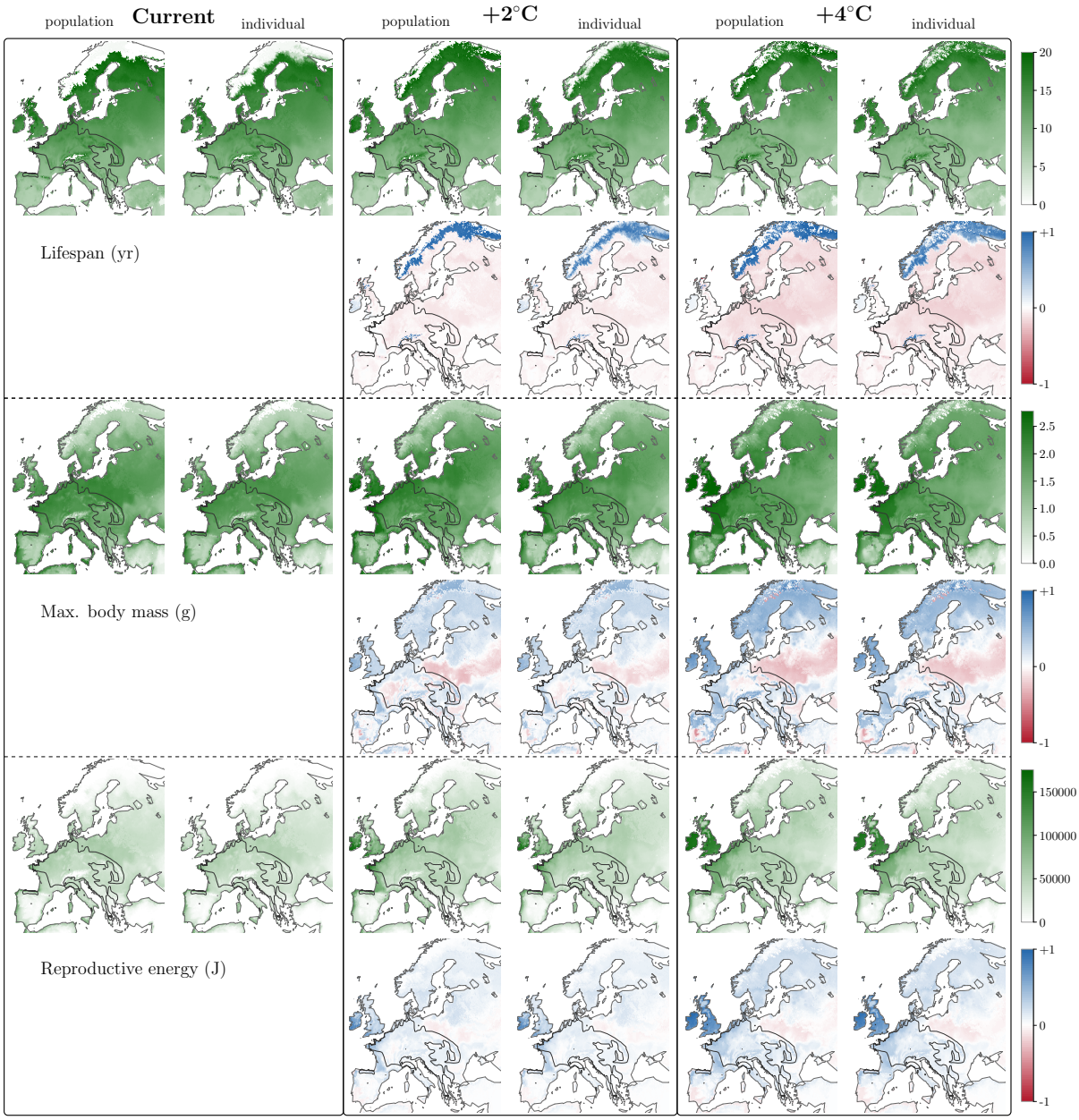


Figure S3.5: Comparison of population-average and mean individual-level predictions for lifespan, maximum body mass and reproductive energy for males under current, +2 °C and +4 °C climate scenarios. For each metric, the top row shows the raw predicted values for all six parameterisation–climate combinations on a shared scale. At the bottom, we show the change relative to current climates for the warming scenarios ($P_{\text{future}} - P_{\text{current}}$) on a normalised scale $[-1, 1]$ (*blue* indicates decrease, while *red* indicates increase). The black outline delimits the current distribution of *I. alpestris* (IUCN 2021).

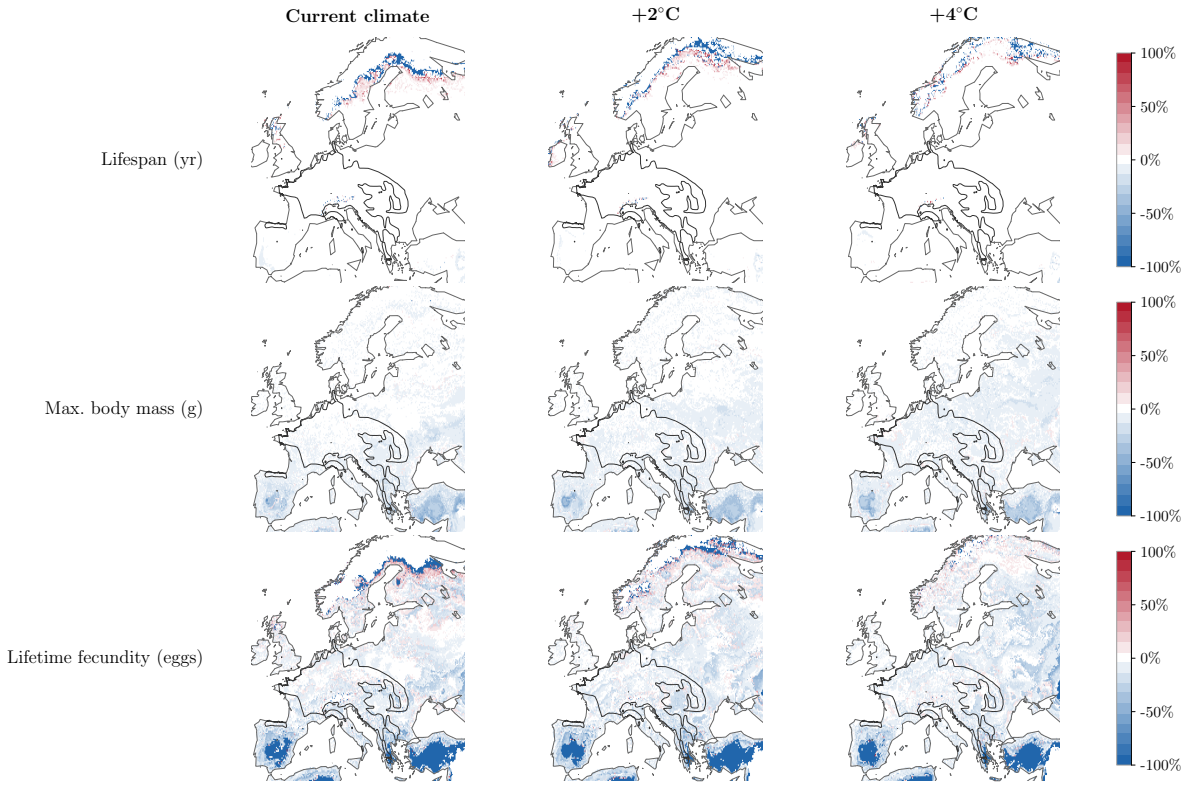


Figure S3.6: Relative bias (%) between population-average and individual-level predictions for females across current, +2 °C and +4 °C scenarios. Relative bias is computed as $R = (P_{\text{pop}} - \overline{P}_{\text{ind}})/|\overline{P}_{\text{ind}}| \times 100$ for each grid cell. Blue indicates underestimation by the population-average model ($R < 0$); red indicates overestimation ($R > 0$); white indicates close agreement ($R \approx 0$).

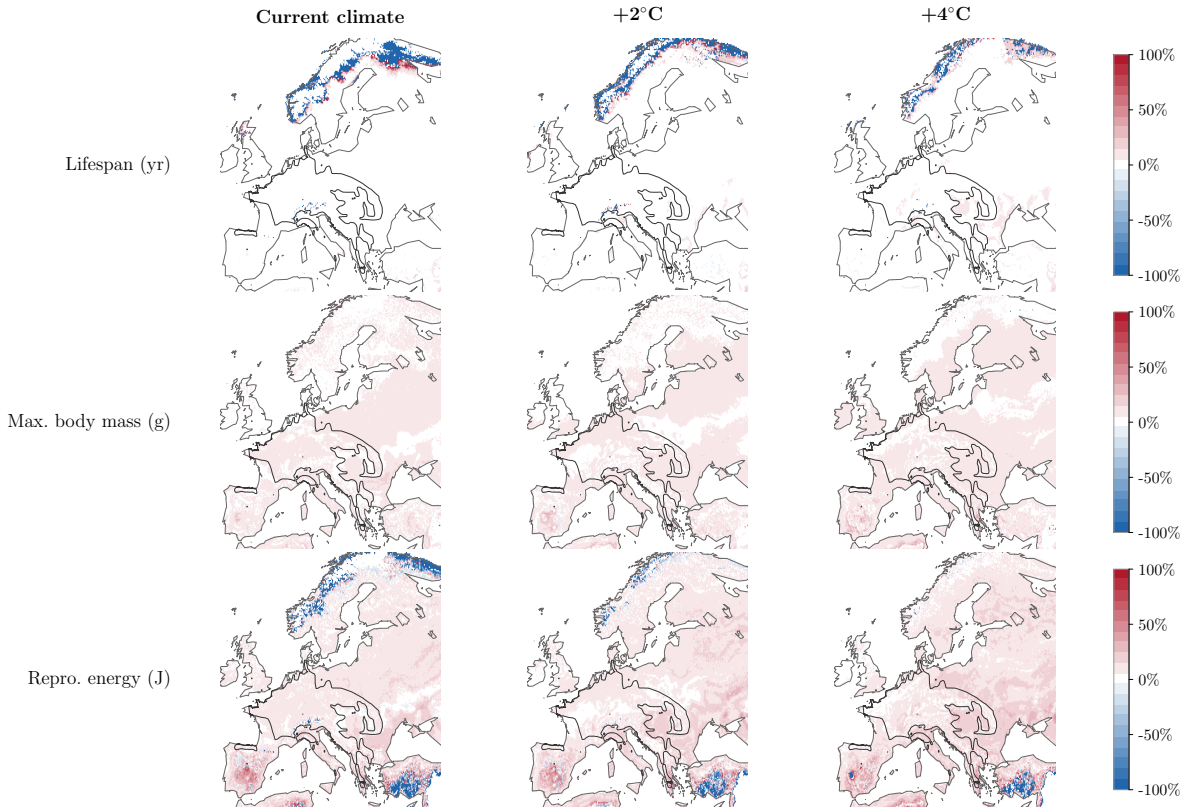


Figure S3.7: Relative bias (%) between population-average and individual-level predictions for males, as in **Fig. S3.6** but showing lifespan, maximum body mass and reproductive energy.

5 References

IUCN (2021). *Ichthyosaura Alpestris: IUCN SSC Amphibian Specialist Group: The IUCN Red List of Threatened Species 2022: E.T59472A89702213*. International Union for Conservation of Nature.